

Non-Functional Requirements

ID	Non-Functional Requirement	Category	Metric / Test	Rationale
Performance				
NFR-01	The system shall support a minimum of 500 concurrent users under normal operating conditions, and shall remain stable under peak loads of up to 1,000 concurrent users without exceeding the response time threshold defined in NFR-02.	Performance	Load test at 500 and 1,000 concurrent users; response time must remain within 2 seconds	Scalability for agency operations
NFR-02	The system shall respond to common user actions within 2 seconds under normal load conditions.	Performance	Measured via response time testing under normal load; 95% of requests must complete within 2 seconds	Responsiveness and usability
Security				
NFR-03	The system shall protect authentication credentials, sensitive customer data, and financial records through industry-standard encryption for data in transit and at rest, ensuring no user can access data beyond their assigned role and permissions.	Security	Verified via security audit; all data transmissions use TLS; stored sensitive data is encrypted at rest	Data protection and privacy
NFR-04	The system shall enforce role-based access control, ensuring that each user type can only access the functions and data appropriate to their assigned role.	Security	Verified by access testing per role; unauthorized access attempts must return access denied	Access control and authorization
NFR-05	Audit log entries shall not be editable or deletable by any user, including administrators, ensuring a tamper-proof record of all significant system actions.	Security	Verified by attempting edit and delete operations on log entries across all roles; all attempts must fail	Tamper-proof audit trail
Availability & Reliability				
NFR-06	The system shall maintain at least 99% availability during business hours, with planned maintenance scheduled outside of peak operational periods.	Availability	Measured via uptime monitoring; maximum allowable downtime during business hours is 1.68 hours per week	Operational continuity
NFR-07	The system shall maintain consistency across all interrelated records including booking status, payment status, package capacity, traveler information, and financial adjustments, such that a booking confirmation always triggers a corresponding capacity update and payment status record within the same operation.	Reliability	Verified via integration testing; no booking confirmation may exist without a matching capacity and payment record update	Data integrity across modules
NFR-08	The system shall perform automated backups of all operational data at least once every 24 hours, and shall be capable of restoring to the most recent backup state within 4 hours of a system failure.	Reliability	Verified via backup logs and recovery drill; restoration must complete within 4 hours of initiated recovery	Business continuity and disaster recovery
Usability				
NFR-09	A first-time user shall be able to complete a standard booking workflow without formal training and within 10 minutes of their first interaction with the system.	Usability	Verified via usability testing with first-time users; 80% must complete the	Ease of use for customers

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			booking workflow within 10 minutes unaided	
NFR-10	The system shall display a confirmation or error message within 2 seconds of every user-initiated action, covering success, failure, and validation states.	Usability	Verified via functional testing; every action must produce a visible feedback message within 2 seconds	User clarity and feedback
NFR-11	All system modules shall use the same navigation structure, button placement conventions, and terminology, ensuring a consistent interface experience across the entire system.	Usability	Verified via UI review against interface style guide; deviations in navigation or terminology constitute a defect	Interface consistency
NFR-12	The system shall support display of interface content in the user's preferred language or detected device language, with English as the default language.	Usability	Verified by switching device language settings; interface must reflect the selected language within one page load	Localization and accessibility
Compliance & Legal				
NFR-13	The system shall comply with applicable legal and regulatory requirements including GDPR or equivalent data privacy legislation, financial record-keeping regulations applicable to travel agencies, and consumer protection laws relevant to the operating jurisdiction.	Compliance	Verified via compliance review against applicable legislation; data handling practices must be documented and auditable	Legal alignment and risk management
NFR-14	The system shall retain booking records and financial data for a minimum of 7 years, and uploaded customer documents for a minimum of 2 years following trip completion, after which records may be archived or purged according to a defined retention policy.	Compliance	Verified via data retention audit; records must remain retrievable for the defined minimum periods	Legal record-keeping obligations
Constraints & Extensibility				
NFR-15	The system shall enforce file size and format constraints on customer document uploads, accepting only PDF, JPEG, and PNG formats with a maximum file size of 10MB per uploaded document.	Constraint	Verified via upload testing; files exceeding 10MB or outside accepted formats must be rejected with a clear error message	Storage management and compatibility
NFR-16	New integration modules shall be addable to the system without modifying existing core system components, supporting future integration with external services such as payment gateways, partner booking platforms, and government travel advisory systems.	Extensibility	Verified via architectural review; addition of a new integration module must not require changes to existing core modules	Future-proofing and scalability